EXPLORATORY DATA ANALYSIS

Titanic Passenger Dataset

Internship Task 5 — Data Analysis Report

Objective	Explore the dataset using descriptive statistics and visualizations to identify data-quality issues, distributions, and relationships.
Tools used	Python, Pandas, Matplotlib, Seaborn
Dataset size	891 passenger records and 12 columns, as shown by the notebook output.
Scope	Missing-value handling, frequency counts, pairplot, correlation heatmap, and plots for survival, age, fare, and passenger class.

This report is based on the code, displayed outputs, and charts saved in the submitted notebook. Findings are described at the level supported by those outputs; no predictive model was built.

1. Executive Summary

This analysis examines passenger information from the Titanic dataset. The notebook uses Pandas to inspect and prepare the data, then uses Matplotlib and Seaborn to explore distributions and relationships. The dataset contains 891 records. Survival is coded as 0 for did not survive and 1 for survived.

Measure	Observed result
Total passenger records	891
Survived	342 (38.4%)
Did not survive	549 (61.6%)
Male passengers	577
Female passengers	314
Age values missing before filling	177 (19.9%)
Cabin values missing	687 (77.1%)
Embarked values missing before filling	2 (0.2%)

Main observations: The notebook's calculations show a substantially higher observed survival rate among female passengers than male passengers. The passenger-class chart shows the highest survival rate for first class and the lowest for third class. Age is concentrated around young-adult ages, while the fare boxplot shows a long upper tail and several high-fare outliers.

Important limitation: These are descriptive associations in this dataset. They do not establish that a characteristic alone caused survival or death.

2. Dataset Overview

The notebook loads a Titanic training dataset using `pd.read_csv('/content/train.csv')`. The `df.info()` output shows 891 rows, 12 columns, and approximately 83.7 KB of memory usage. The columns cover survival outcome, passenger class, name, sex, age, family counts, ticket, fare, cabin, and embarkation port.

The notebook output identifies five integer columns, two floating-point columns, and five object-type columns. Age and Fare are numeric continuous-style fields; Pclass, SibSp, Parch, and Survived are integer-coded variables. The meaning of the codes should be kept in mind when interpreting correlations.

3. Data Quality and Preparation

The notebook uses `df.isnull().sum()` to inspect missing data and then applies two filling steps:

• **Age:** 177 missing values are filled with the mean age. The displayed values indicate a mean of approximately 29.7 years.

• **Embarked:** 2 missing values are filled with the most frequent category (mode).

• **Cabin:** 687 values are missing. The submitted notebook does not fill or otherwise resolve this column's missing values.

Using the mean and mode allows the notebook to continue with fewer missing values in Age and Embarked. However, mean imputation can make the age distribution more concentrated around the mean. Cabin missingness is substantial and should be considered if that column is used in later analysis. The notebook does not show a duplicate-record check or a systematic outlier-treatment step.

4. Exploratory Visualizations and Findings

4.1 Pairplot: selected passenger variables

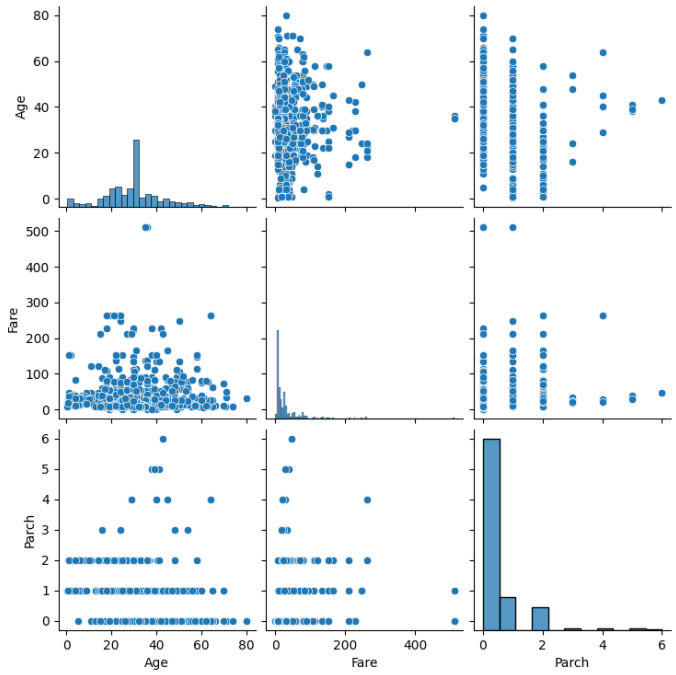


Figure 1. Pairplot generated in the notebook for Age, Sex, Fare, and Parch. The displayed axes emphasize the numeric variables Age, Fare, and Parch.

The diagonal panels show individual-variable distributions, while off-diagonal panels show pairwise scatterplots. Age values cluster most heavily in the adult range, and Fare has a concentrated lower range with a few much higher values. Parch is discrete, so horizontal bands and overlapping points are expected. The chart is useful for screening patterns, but it does not by itself establish a strong relationship between fare and family size.

4.2 Correlation heatmap

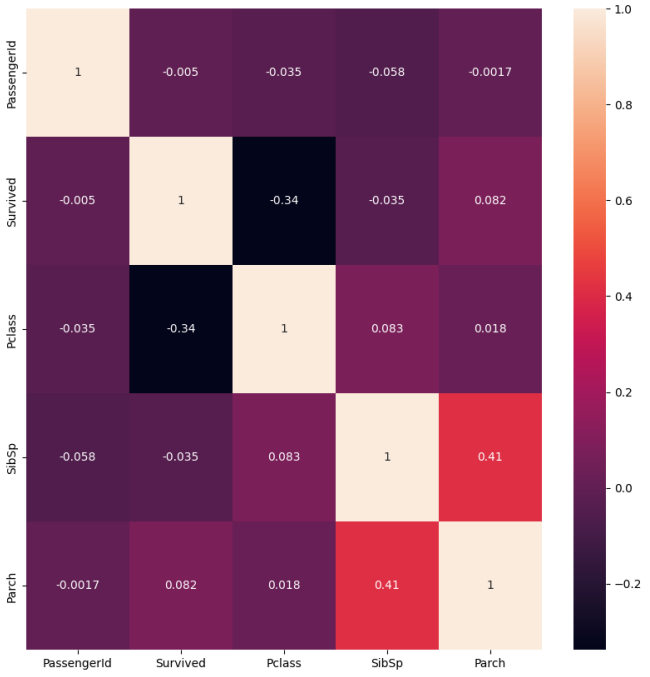


Figure 2. Correlation heatmap for the integer-typed columns selected by the notebook.

The strongest visible positive association in the heatmap is between SibSp and Parch (about 0.41), suggesting that passengers travelling with siblings/spouses also tended, to some extent, to travel with parents/children. Survived and Pclass have a negative correlation of about -0.34. Because a smaller Pclass number represents a higher passenger class, this is consistent with higher-class passengers having a higher survival rate in this dataset. Correlation coefficients reflect linear association and do not prove causation. PassengerId is an identifier and its correlations are not substantively meaningful.

4.3 Survival rate by sex

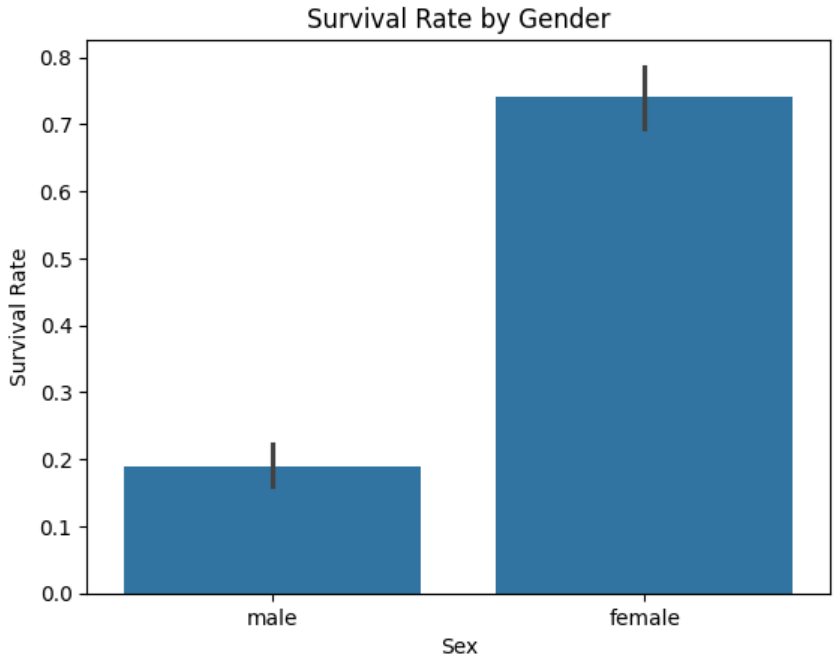


Figure 3. Mean of Survived by Sex; since Survived is coded 0/1, the bar height represents the observed survival proportion.

The notebook calculates a survival rate of 74.20% for female passengers and 18.89% for male passengers. This is one of the clearest differences in the analysis. In this dataset, the observed survival proportion among women was about four times the proportion among men. This is a descriptive comparison and does not control for other variables such as passenger class or age.

4.4 Ticket fare and Parch

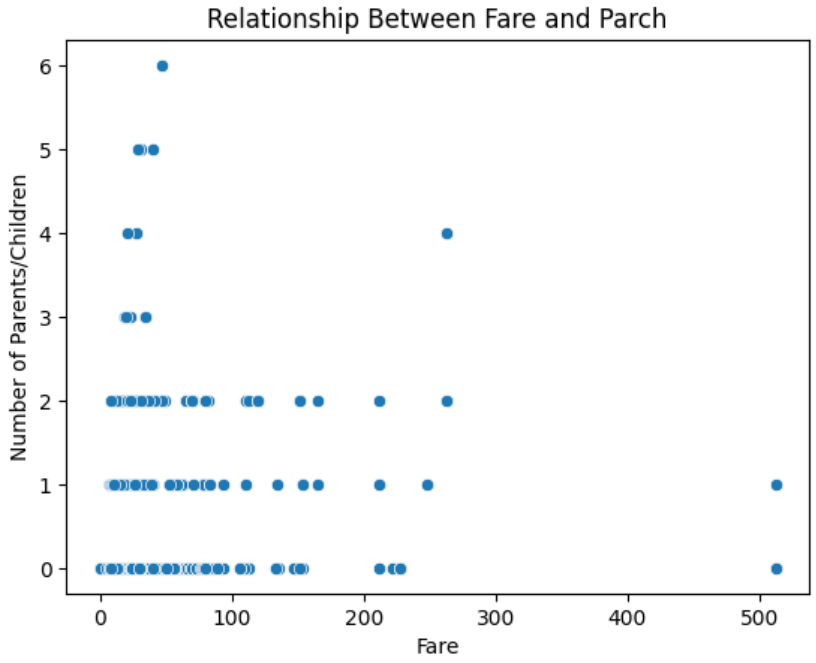


Figure 4. Scatterplot of Fare against Parch.

Most plotted passengers are concentrated at lower fares, and Parch values appear in discrete horizontal bands. A few passengers paid fares above 200, including a point above 500. The scatterplot does not show a simple linear pattern between fare and Parch; the high-fare points are relatively sparse. Because many observations overlap, this plot should be interpreted as a broad exploratory view rather than a precise measure of association.

4.5 Passenger age distribution

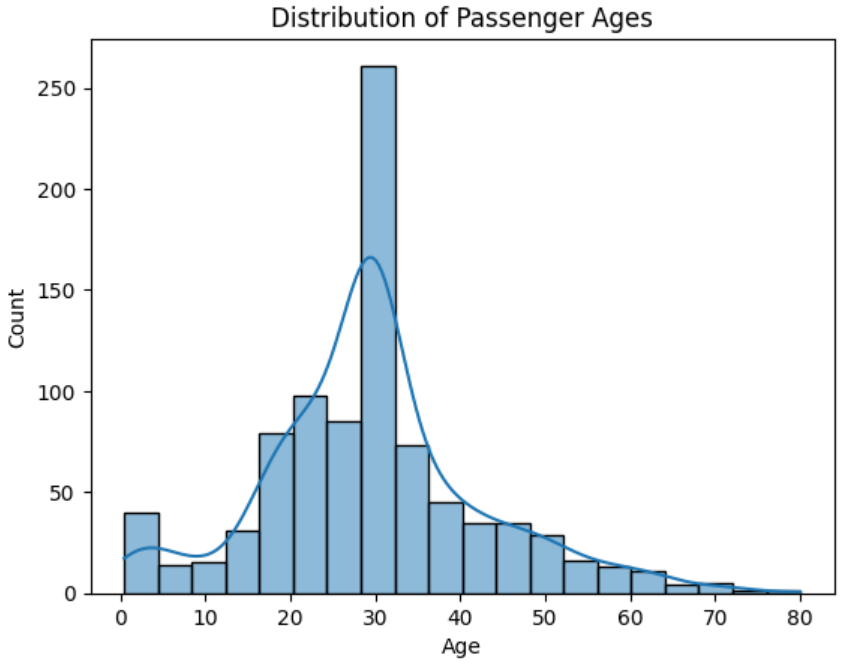


Figure 5. Histogram and KDE curve for Age, using 20 bins.

The histogram shows the largest concentration of ages around the late twenties to early thirties. Counts decline toward older ages, producing a right tail. Children are present, but the bulk of passengers in the displayed distribution are adults. Since missing ages were filled with the mean before this plot was generated, the filled values may

contribute to the concentration near the mean.

4.6 Survival rate by passenger class

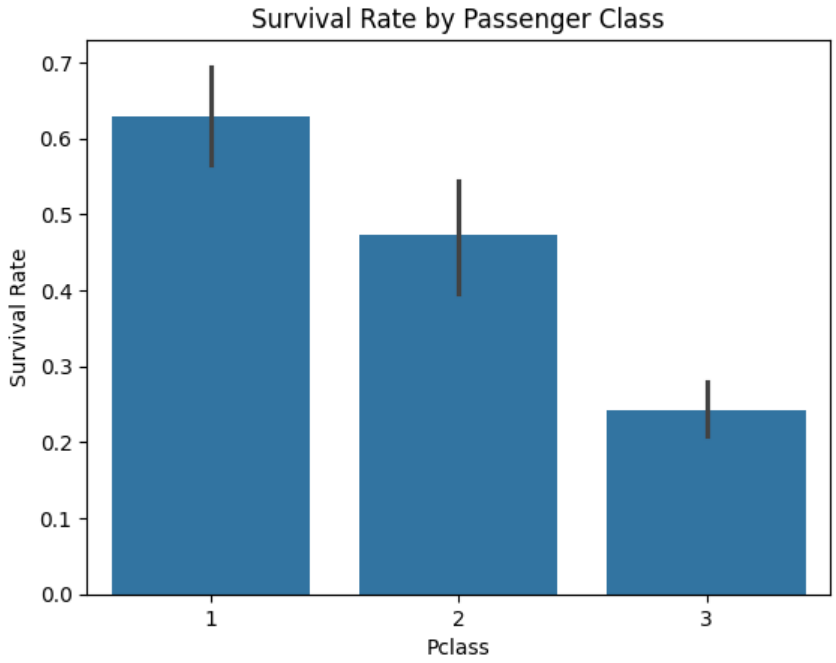


Figure 6. Mean survival by Pclass. Error bars are displayed by Seaborn's default barplot estimator.

The chart shows a clear class gradient: first-class passengers have the highest observed survival rate (approximately 63%), second-class passengers are next (approximately 47%), and third-class passengers have the lowest (approximately 24%). This pattern aligns with the negative correlation between Pclass and Survived shown in the heatmap. The chart compares groups but does not isolate the effect of class from other characteristics.

4.7 Distribution of ticket fares

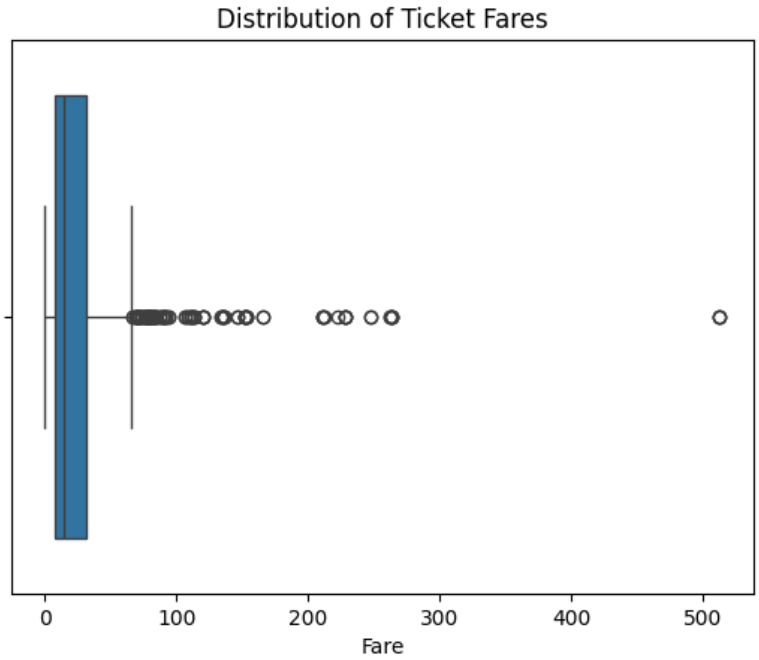


Figure 7. Boxplot of Fare. Individual points beyond the whiskers indicate potential outliers.

The fare distribution is strongly right-skewed: most fares fall in the lower range, while a smaller number of passengers paid much higher fares. Several points extend beyond the upper whisker, with one especially high value around 512. These are potential outliers, not automatically errors. They may reflect differences in ticket type, passenger grouping, or class and would need checking before any values were removed.

5. Key Findings

Survival overall: 342 of 891 passengers survived (38.4%); 549 (61.6%) did not.

Sex and survival: Female survival was 74.20%, compared with 18.89% for male passengers.

Passenger class: First class had the highest observed survival rate; third class had the lowest.

Missing data: Age had 177 missing values, Cabin had 687, and Embarked had 2 before the notebook's filling steps.

Age pattern: Passenger ages were concentrated around adult ages, with fewer older passengers.

Fare pattern: Ticket fares were right-skewed and included several high-fare potential outliers.

Family-count association: SibSp and Parch had a moderate positive correlation of approximately 0.41.

6. Conclusion

This EDA used basic data inspection, missing-value handling, frequency counts, a pairplot, a correlation heatmap, and several focused visualizations. The most prominent patterns in the saved outputs are the difference in survival proportions by sex, the survival gradient across passenger classes, and the right-skewed fare distribution. The heatmap also highlights an association between the two family-count variables, SibSp and Parch.

The notebook provides a useful first overview of the Titanic data, but it is exploratory rather than predictive. Further work could compare survival across sex and class together, inspect the Cabin column's missingness, check for duplicate records, and document summary statistics such as medians and quartiles. Any further conclusions should be based on re-running the analysis on the same source data and validating the relevant counts.

7. Tools and Methods Used

Tool / method	Purpose in this notebook
Pandas: info(), isnull().sum()	Inspect column types, dataset size, and missing values.
Pandas: fillna(), value_counts()	Fill missing Age/Embarked values and count categories.
Seaborn: pairplot()	Explore pairwise distributions and possible associations.
Seaborn: heatmap()	Visualize correlations among integer-typed columns.
Seaborn: barplot(), histplot(), boxplot(), scatterplot()	Compare survival proportions and inspect distributions/relationships.
Matplotlib: titles and show()	Display and label the plots.

Reproducibility note: the notebook reads `/content/train.csv`, which is a Google Colab path. Include the dataset in the repository or update the notebook to use a documented relative path so another person can run it.

Appendix: Report scope

This report was prepared from the submitted `EDA_Titanic.ipynb` and its saved outputs. Values explicitly shown in notebook outputs are reported directly. Class survival percentages are described approximately from the saved bar chart; they should be regenerated from the dataset if exact class-specific percentages are required.